Title: Creating Arrays

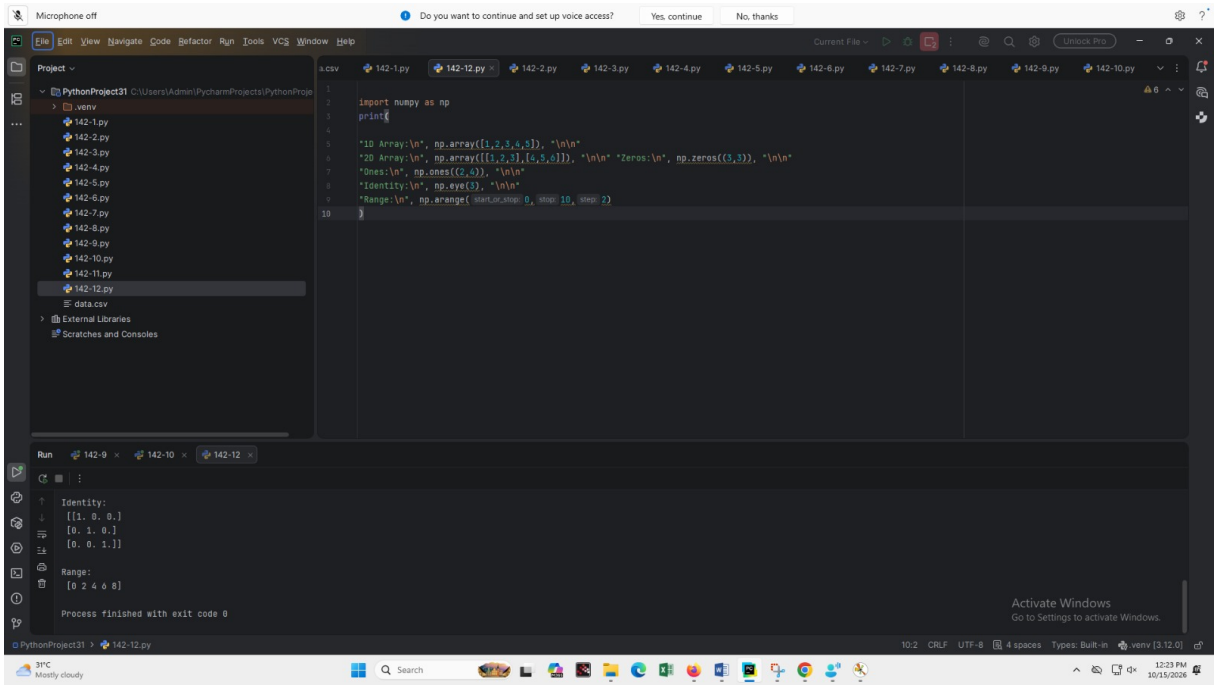
Aim:

To create 1-D, 2-D and 3-D NumPy arrays and display their shape, size, dimension and data type.

Algorithm:

1. Start.
2. Import the NumPy library (`import numpy as np`).
3. Create 1-D, 2-D and 3-D arrays using `np.array()`.
4. Display each array's `shape`, `ndim`, `size` and `dtype`.
5. Stop.

Program:



Title: Array Operations

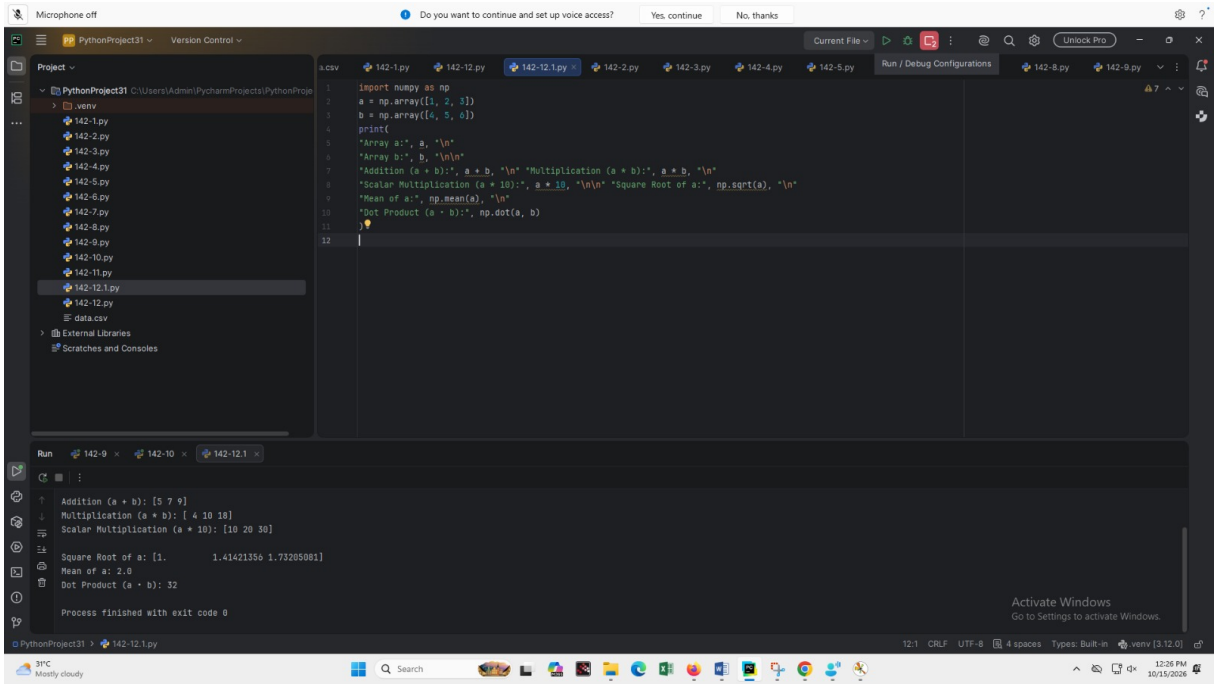
Aim:

To perform matrix addition, multiplication, transpose, determinant and inverse using NumPy.

Algorithm:

1. Start.
2. Import NumPy.
3. Create two square matrices A and B.
4. Compute A + B, and A x B using `np.dot()` or `@`.
5. Compute the transpose using `A.T`.
6. Compute the determinant using `np.linalg.det(A)`.
7. Compute the inverse using `np.linalg.inv(A)` if the determinant is non-zero.
8. Display the results.
9. Stop.

Program:



Title: Indexing and Slicing

Aim:

To access elements, rows, columns and sub-arrays of a NumPy array using indexing and slicing.

Algorithm:

1. Start.
2. Import NumPy and create a 2-D array.
3. Access a single element using `arr[row, col]`.
4. Extract rows, columns and sub-arrays using slicing (`arr[start:stop, start:stop]`).
5. Display the results.
6. Stop.

Program:

